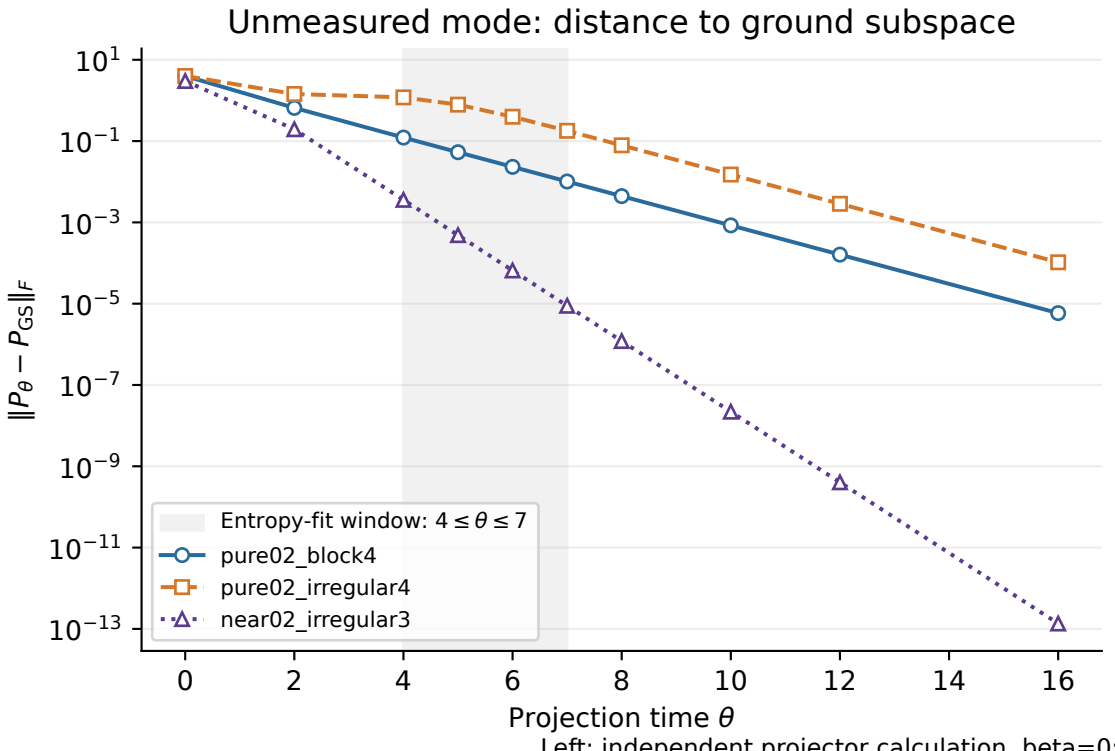
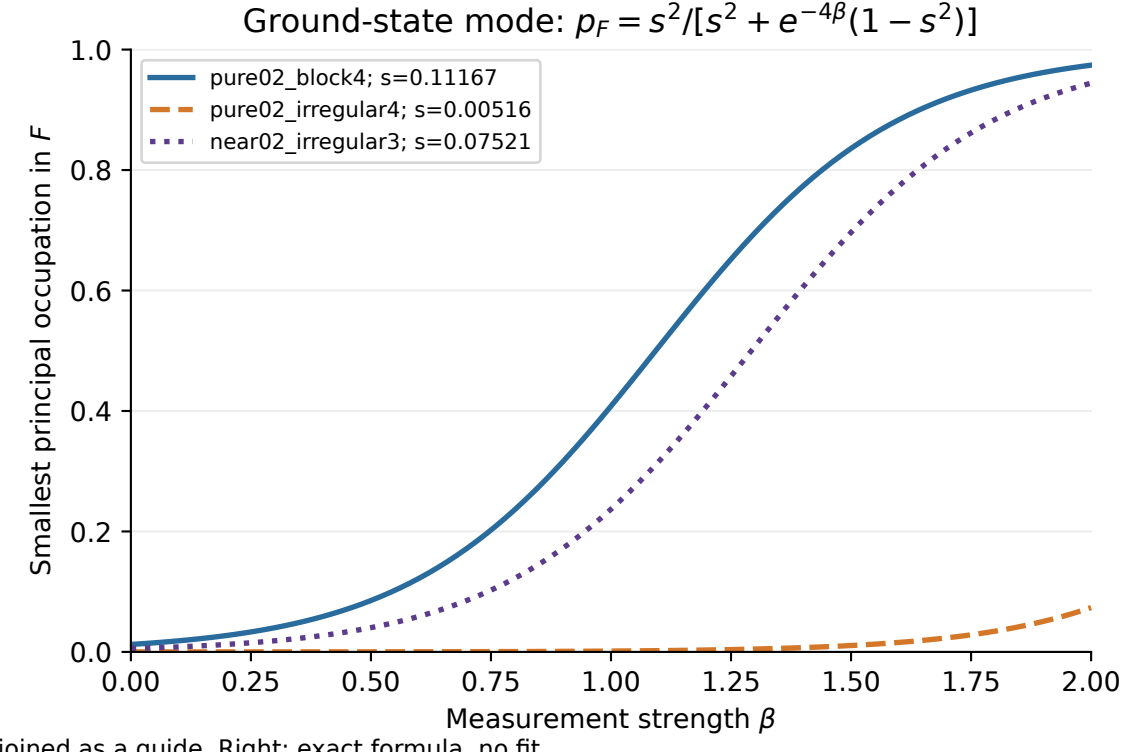
Single-mode mechanism diagnostic: $k = (\pi, 0)$ in supercell reciprocal coordinates
All three modes are gapped; the two m=4 patterns share the same energy gap





Left: independent projector calculation, beta=0; points joined as a guide. Right: exact formula, no fit.
F = initially occupied sites; s = minimum ground/initial overlap singular value at this fixed k.
Neither panel is an entropy or a scaling coefficient; fixed-k s is not the minimum over the Brillouin zone.